

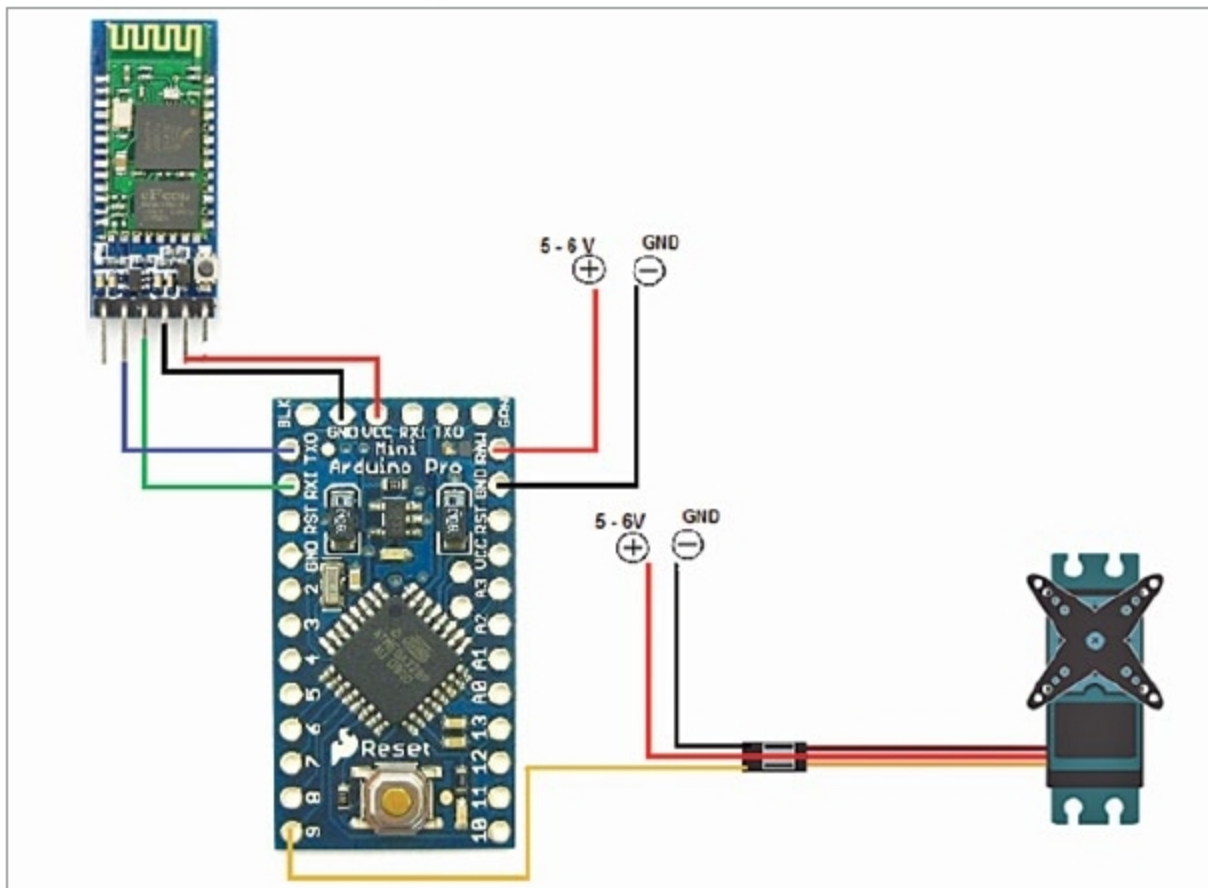


Fig. 5: Circuit connections



Fig. 6: Lock used by author



Fig. 7: Lock lever

module. Now, define the pin for the servo motor that you will use. Refer the code given in Fig. 1.

Next, create a loop function where you store the device ID sent by Bluetooth in reads string. Then, create an if condition that checks the device ID



Fig. 8: Servo fin attached to the lever



Fig. 9: Three wires from the servo inside the lock

sent from Bluetooth. If the fingerprint matches in Android app, the app will send the device ID to Arduino. Initially, the servo is in lock position. If device ID set in Arduino and ID sent by app match, the program moves the servo to unlock the lock.

But if the app reads a wrong fingerprint, the servo remains in lock position. Refer Fig. 2 for details.

Arduino Connections Table 2

Arduino	Components
Rx	Bluetooth Tx
Tx	Bluetooth Rx
5V	Bluetooth 5V
GND	Bluetooth GND
GND	Servo motor GND
Pin 9	Servo motor signal wire (yellow coloured)

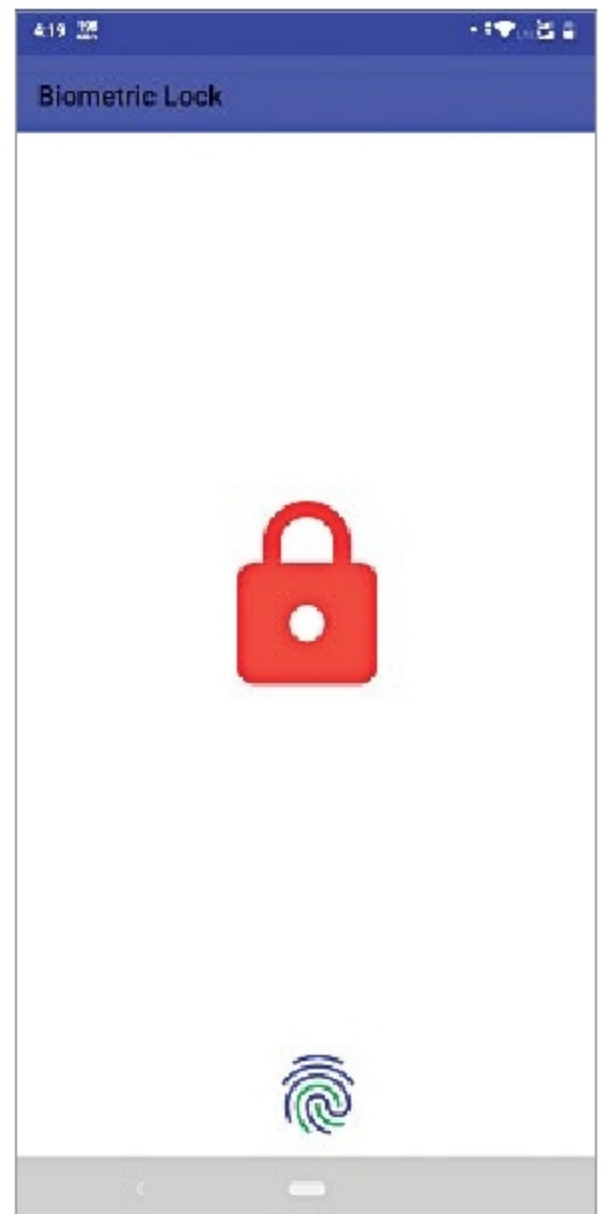


Fig. 10: App layout

Development of Android app

There are many tools available for developing the Android app, including Android Studio, Kodular and MIT App Inventor. We have used Kodular. It has different code blocks through which you can make the app with different combinations of these blocks. It is easy and saves time during coding.

Open any Web browser and search for Kodular. To start making the app, open Kodular and add the following components in the layout (Fig. 3):

- Bluetooth client
- Fingerprint
- List view
- Image button